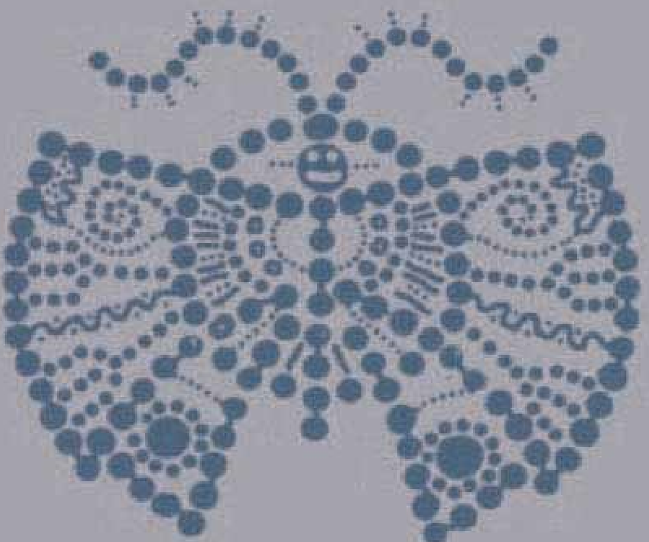
Springer Series in Operations Research

Jorge Nocedal
Stephen J. Wright

Numerical Optimization

Second Edition



Springer

Springer Series in Operations Research and Financial Engineering

Springer Series in Operation Research and Financial Engineering

Altiok: Performance Analysis of Manufacturing Systems
Birge and Louveaux: Introduction to Stochastic Programming
Bonnans and Shapiro: Perturbation Analysis of Optimization Problems
Bramel, Chen, and Simchi-Levi: The Logic of Logistics: Theory, Algorithms, and Applications for Logistics and Supply Chain Management (second edition)
Dantzig and Thapa: Linear Programming 1: Introduction
Dantzig and Thapa: Linear Programming 2: Theory and Extensions
Drezner (Editor): Facility Location: A Survey of Applications and Methods
Facchinei and Pang: Finite-Dimensional Variational Inequalities and Complementarity Problems, Volume I
Facchinei and Pang: Finite-Dimensional Variational Inequalities and Complementarity Problems, Volume II
Fishman: Discrete-Event Simulation: Modeling, Programming, and Analysis
Fishman: Monte Carlo: Concepts, Algorithms, and Applications
Haas: Stochastic Petri Nets: Modeling, Stability, Simulation
Klamroth: Single-Facility Location Problems with Barriers
Muckstadt: Analysis and Algorithms for Service Parts Supply Chains
Nocedal and Wright: Numerical Optimization
Olson: Decision Aids for Selection Problems
Pinedo: Planning and Scheduling in Manufacturing and Services
Pochet and Wolsey: Production Planning by Mixed Integer Programming
Whitt: Stochastic-Process Limits: An Introduction to Stochastic-Process Limits and Their Application to Queues
Yao (Editor): Stochastic Modeling and Analysis of Manufacturing Systems
Yao and Zheng: Dynamic Control of Quality in Production-Inventory Systems: Coordination and Optimization
Yeung and Petrosyan: Cooperative Stochastic Differential Games

Jorge Nocedal Stephen J. Wright

Numerical Optimization

Second Edition

Springer

Jorge Nocedal
EECS Department
Northwestern University
Evanston, IL 60208-3118
USA
nocedal@eecs.northwestern.edu

Stephen J. Wright
Computer Sciences Department
University of Wisconsin
1210 West Dayton Street
Madison, WI 53706–1613
USA
swright@cs.wisc.edu

Mathematics Subject Classification (2000): 90B30, 90C11, 90-01, 90-02

Library of Congress Control Number: 2006923897

ISBN-10: 0-387-30303-0 ISBN-13: 978-0387-30303-1

Printed on acid-free paper.

Printed in the United States of America. (TB/HAM)

9 8 7 6 5 4 3 2 1

springer.com

To Sue, Isabel and Martin
and
To Mum and Dad

Contents

Preface

This is a book for people interested in solving optimization problems. Because of the wide (and growing) use of optimization in science, engineering, economics, and industry, it is essential for students and practitioners alike to develop an understanding of optimization algorithms. Knowledge of the capabilities and limitations of these algorithms leads to a better understanding of their impact on various applications, and points the way to future research on improving and extending optimization algorithms and software. Our goal in this book is to give a comprehensive description of the most powerful, state-of-the-art, techniques for solving continuous optimization problems. By presenting the motivating ideas for each algorithm, we try to stimulate the reader's intuition and make the technical details easier to follow. Formal mathematical requirements are kept to a minimum.

Because of our focus on continuous problems, we have omitted discussion of important optimization topics such as discrete and stochastic optimization. However, there are a great many applications that can be formulated as continuous optimization problems; for instance,

finding the optimal trajectory for an aircraft or a robot arm;

identifying the seismic properties of a piece of the earth's crust by fitting a model of the region under study to a set of readings from a network of recording stations;

designing a portfolio of investments to maximize expected return while maintaining an acceptable level of risk;

controlling a chemical process or a mechanical device to optimize performance or meet standards of robustness;

computing the optimal shape of an automobile or aircraft component.

Every year optimization algorithms are being called on to handle problems that are much larger and complex than in the past. Accordingly, the book emphasizes large-scale optimization techniques, such as interior-point methods, inexact Newton methods, limited-memory methods, and the role of partially separable functions and automatic differentiation. It treats important topics such as trust-region methods and sequential quadratic programming more thoroughly than existing texts, and includes comprehensive discussion of such "core curriculum" topics as constrained optimization theory, Newton and quasi-Newton methods, nonlinear least squares and nonlinear equations, the simplex method, and penalty and barrier methods for nonlinear programming.

The Audience

We intend that this book will be used in graduate-level courses in optimization, as offered in engineering, operations research, computer science, and mathematics departments. There is enough material here for a two-semester (or three-quarter) sequence of courses. We hope, too, that this book will be used by practitioners in engineering, basic science, and industry, and our presentation style is intended to facilitate self-study. Since the book treats a number of new algorithms and ideas that have not been described in earlier textbooks, we hope that this book will also be a useful reference for optimization researchers.

Prerequisites for this book include some knowledge of linear algebra (including numerical linear algebra) and the standard sequence of calculus courses. To make the book as self-contained as possible, we have summarized much of the relevant material from these areas in the Appendix. Our experience in teaching engineering students has shown us that the material is best assimilated when combined with computer programming projects in which the student gains a good feeling for the algorithms—their complexity, memory demands, and elegance—and for the applications. In most chapters we provide simple computer exercises that require only minimal programming proficiency.

Emphasis and Writing Style

We have used a conversational style to motivate the ideas and present the numerical algorithms. Rather than being as concise as possible, our aim is to make the discussion flow in a natural way. As a result, the book is comparatively long, but we believe that it can be read relatively rapidly. The instructor can assign substantial reading assignments from the text and focus in class only on the main ideas.

A typical chapter begins with a nonrigorous discussion of the topic at hand, including figures and diagrams and excluding technical details as far as possible. In subsequent sections, the algorithms are motivated and discussed, and then stated explicitly. The major theoretical results are stated, and in many cases proved, in a rigorous fashion. These proofs can be skipped by readers who wish to avoid technical details.

The practice of optimization depends not only on efficient and robust algorithms, but also on good modeling techniques, careful interpretation of results, and user-friendly software. In this book we discuss the various aspects of the optimization process—modeling, optimality conditions, algorithms, implementation, and interpretation of results—but not with equal weight. Examples throughout the book show how practical problems are formulated as optimization problems, but our treatment of modeling is light and serves mainly to set the stage for algorithmic developments. We refer the reader to Dantzig [86] and Fourer, Gay, and Kernighan [112] for more comprehensive discussion of this issue. Our treatment of optimality conditions is thorough but not exhaustive; some concepts are discussed more extensively in Mangasarian [198] and Clarke [62]. As mentioned above, we are quite comprehensive in discussing optimization algorithms.

Topics Not Covered

We omit some important topics, such as network optimization, integer programming, stochastic programming, nonsmooth optimization, and global optimization. Network and integer optimization are described in some excellent texts: for instance, Ahuja, Magnanti, and Orlin [1] in the case of network optimization and Nemhauser and Wolsey [224], Papadimitriou and Steiglitz [235], and Wolsey [312] in the case of integer programming. Books on stochastic optimization are only now appearing; we mention those of Kall and Wallace [174], Birge and Louveaux [22]. Nonsmooth optimization comes in many flavors. The relatively simple structures that arise in robust data fitting (which is sometimes based on the ℓ_1 norm) are treated by Osborne [232] and Fletcher [101]. The latter book also discusses algorithms for nonsmooth penalty functions that arise in constrained optimization; we discuss these briefly, too, in Chapter 18. A more analytical treatment of nonsmooth optimization is given by Hiriart-Urruty and Lemaréchal [170]. We omit detailed treatment of some important topics that are the focus of intense current research, including interior-point methods for nonlinear programming and algorithms for complementarity problems.

Additional Resource

The material in the book is complemented by an online resource called the NEOS Guide, which can be found on the World-Wide Web at

`http://www.mcs.anl.gov/otc/Guide/`

The Guide contains information about most areas of optimization, and presents a number of case studies that describe applications of various optimization algorithms to real-world problems such as portfolio optimization and optimal dieting. Some of this material is interactive in nature and has been used extensively for class exercises.

For the most part, we have omitted detailed discussions of specific software packages, and refer the reader to Moré and Wright [217] or to the Software Guide section of the NEOS Guide, which can be found at

`http://www.mcs.anl.gov/otc/Guide/SoftwareGuide/`

Users of optimization software refer in great numbers to this web site, which is being constantly updated to reflect new packages and changes to existing software.

Acknowledgments

We are most grateful to the following colleagues for their input and feedback on various sections of this work: Chris Bischof, Richard Byrd, George Corliss, Bob Fourer, David Gay, Jean-Charles Gilbert, Phillip Gill, Jean-Pierre Goux, Don Goldfarb, Nick Gould, Andreas Griewank, Matthias Heinkenschloss, Marcelo Marazzi, Hans Mittelmann, Jorge Moré, Will Naylor, Michael Overton, Bob Plemmons, Hugo Scolnik, David Stewart, Philippe Toint, Luis Vicente, Andreas Wächter, and Ya-xiang Yuan. We thank Guanghui Liu, who provided help with many of the exercises, and Jill Lavelle who assisted us in preparing the figures. We also express our gratitude to our sponsors at the Department of Energy and the National Science Foundation, who have strongly supported our research efforts in optimization over the years.

One of us (JN) would like to express his deep gratitude to Richard Byrd, who has taught him so much about optimization and who has helped him in very many ways throughout the course of his career.

Final Remark

In the preface to his 1987 book [101], Roger Fletcher described the field of optimization as a "fascinating blend of theory and computation, heuristics and rigor." The ever-growing realm of applications and the explosion in computing power is driving optimization research in new and exciting directions, and the ingredients identified by Fletcher will continue to play important roles for many years to come.

Jorge Nocedal
Evanston, IL

Stephen J. Wright
Argonne, IL

Preface to the Second Edition

During the six years since the first edition of this book appeared, the field of continuous optimization has continued to grow and evolve. This new edition reflects a better understanding of constrained optimization at both the algorithmic and theoretical levels, and of the demands imposed by practical applications. Perhaps most notably, new chapters have been added on two important topics: derivative-free optimization (Chapter 9) and interior-point methods for nonlinear programming (Chapter 19). The former topic has proved to be of great interest in applications, while the latter topic has come into its own in recent years and now forms the basis of successful codes for nonlinear programming.

Apart from the new chapters, we have revised and updated throughout the book, de-emphasizing or omitting less important topics, enhancing the treatment of subjects of evident interest, and adding new material in many places. The first part (unconstrained optimization) has been comprehensively reorganized to improve clarity. Discussion of Newton's method—the touchstone method for unconstrained problems—is distributed more naturally throughout this part rather than being isolated in a single chapter. An expanded discussion of large-scale problems appears in Chapter 7.

Some reorganization has taken place also in the second part (constrained optimization), with material common to sequential quadratic programming and interior-point methods now appearing in the chapter on fundamentals of nonlinear programming algorithms (Chapter 15) and the discussion of primal barrier methods moved to the new interior-point chapter. There is much new material in this part, including a treatment of nonlinear programming duality, an expanded discussion of algorithms for inequality constrained quadratic programming, a discussion of dual simplex and presolving in linear programming, a summary of practical issues in the implementation of interior-point linear programming algorithms, a description of conjugate-gradient methods for quadratic programming, and a discussion of filter methods and nonsmooth penalty methods in nonlinear programming algorithms.

In many chapters we have added a Perspectives and Software section near the end, to place the preceding discussion in context and discuss the state of the art in software. The appendix has been rearranged with some additional topics added, so that it can be used in a more stand-alone fashion to cover some of the mathematical background required for the rest of the book. The exercises have been revised in most chapters. After these many additions, deletions, and changes, the second edition is only slightly longer than the first, reflecting our belief that careful selection of the material to include and exclude is an important responsibility for authors of books of this type.

A manual containing solutions for selected problems will be available to bona fide instructors through the publisher. A list of typos will be maintained on the book's web site, which is accessible from the web pages of both authors.

We acknowledge with gratitude the comments and suggestions of many readers of the first edition, who sent corrections to many errors and provided valuable perspectives on the material, which led often to substantial changes. We mention in particular Frank Curtis, Michael Ferris, Andreas Griewank, Jacek Gondzio, Sven Leyffer, Philip Loewen, Rembert Reemtsen, and David Stewart.

Our special thanks goes to Michael Overton, who taught from a draft of the second edition and sent many detailed and excellent suggestions. We also thank colleagues who read various chapters of the new edition carefully during development, including Richard Byrd, Nick Gould, Paul Hovland, Gabo Lopéz-Calva, Long Hei, Katya Scheinberg, Andreas Wächter, and Richard Waltz. We thank Jill Wright for improving some of the figures and for the new cover graphic.

We mentioned in the original preface several areas of optimization that are not covered in this book. During the past six years, this list has only grown longer, as the field has continued to expand in new directions. In this regard, the following areas are particularly noteworthy: optimization problems with complementarity constraints, second-order cone and semidefinite programming, simulation-based optimization, robust optimization, and mixed-integer nonlinear programming. All these areas have seen theoretical and algorithmic advances in recent years, and in many cases developments are being driven by new classes of applications. Although this book does not cover any of these areas directly, it provides a foundation from which they can be studied.

Jorge Nocedal	Stephen J. Wright
Evanston, IL	*Madison, WI*